

Advanced (Carolina Reaper)

- Q1.** A bicycle travels 15 km in 45 minutes. What is its speed in km/h?
(A) 20
(B) 16
(C) 12
(D) 8
(E) 4
- Q2.** A water tank can be filled in 4 hours. How long will it take to fill $\frac{3}{4}$ of the tank?
(A) 3 hours
(B) 2.5 hours
(C) 1.5 hours
(D) 3.5 hours
(E) 2 hours
- Q3.** If y varies directly with x , and $y = 12$ when $x = 4$, what is the value of y when $x = 6$?
(A) 12
(B) 16
(C) 18
(D) 20
(E) 24
- Q4.** A bakery sells 480 loaves of bread per day. If they make a profit of \$0.50 per loaf, what is their daily profit?
(A) 240
(A) 200
(A) 160
(A) 120
(A) 100
- Q5.** A car travels from City A to City B at 60 km/h and returns at 40 km/h. What is its average speed for the round trip?
(A) 45 km/h
(B) 48 km/h
(C) 50 km/h
(D) 52 km/h
(E) 55 km/h
- Q6.** If y varies inversely with x , and $y = 8$ when $x = 2$, what is the value of y when $x = 4$?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 16
- Q7.** A group of friends want to share some candy equally. If they have 48 pieces of candy and there are 8 friends, how many pieces will each friend get?
(A) 4
(B) 5
(C) 6
(D) 7
(E) 8
- Q8.** A book costs \$20. If a 10% discount is applied, how much will you pay for the book?
(A) 18
(A) 19
(A) 20
(A) 21
(A) 22

Open Ended Questions (FRQ)

- Q9.** Tom has been saving money for a new bike and has \$120 saved. He wants to buy a bike that costs \$180. If he saves \$10 per week, how many weeks will it take him to have enough money to buy the bike?
- Q10.** A water tank can hold 3000 liters of water. If 1200 liters of water are already in the tank, what percentage of the tank is filled?

Chef's Tips

- Read each question carefully
- Use a calculator only when necessary
- Show all work for free-response questions